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AI speech-based emotion recognition

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Introduction to Graduation Final Report submitted to the
Department of Electrical and Computer Engineering
in partial fulfillment of the requirements for the
degree of B.Sc. in Computer Engineering

First Semester
2025/2026
Birzeit, Palestine

Abstract

Emotions are important for human communication, and recognizing them from speech enhances human-computer interaction in healthcare, call centers, and virtual assistants. This collaborative project addresses Speech Emotion Recognition (SER) through two complementary approaches: a traditional machine learning baseline using ensemble methods, and a frequency-based clustering approach combined with deep learning.

Baseline Approach: We developed a robust baseline using traditional machine learning methods on the RAVDESS dataset. Feature extraction included MFCCs, pitch, spectral characteristics, and energy features. Our ensemble approach combining SVM and Random Forest achieved 89.49% accuracy on 5 emotions and 83.68% accuracy on the full 8-emotion dataset.

Frequency-Based Clustering Approach: Building on the frequency-based clustering methodology proposed by Barhoumi & BenAyed[1], this approach groups the full 8-emotion RAVDESS dataset by arousal characteristics. High-arousal emotions (angry, fearful, happy, surprised) are separated from low-arousal emotions (calm, disgust, neutral, sad). Specialized CNN-BiLSTM models trained on each cluster achieved 67% accuracy for high-arousal emotions and 59% for low-arousal emotions with dynamic features.

This thesis presents both approaches, demonstrating that traditional ML with well-engineered features achieves higher performance 83.68% compared to deep learning 56% on limited data, while frequency-based clustering provides a principled framework for organizing emotions by acoustic properties.

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List of Abbreviations and Symbols

Abbreviations

AI	Artificial Intelligence
BiLSTM	Bidirectional Long Short-Term Memory
CNN	Convolutional Neural Network
CV	Cross-Validation
F0	Fundamental Frequency
F1	F1-Score
LSTM	Long Short-Term Memory
MFCC	Mel-Frequency Cepstral Coefficient
ML	Machine Learning
pp	Percentage Points
RAVDESS	Ryerson Audio-Visual Database of Emotional Speech and Song
RBF	Radial Basis Function
RF	Random Forest
RMS	Root Mean Square
RNN	Recurrent Neural Network
SER	Speech Emotion Recognition
SVM	Support Vector Machine
ZCR	Zero Crossing Rate

Mathematical Symbols

T	Time dimension (sequence length)
F	Feature dimension
σ	Standard deviation
μ	Mean
C	Regularization parameter (SVM)
γ	Kernel coefficient (SVM)
f_0	Fundamental frequency (pitch)

Chapter 1

Introduction and Literature Review

1.1 Introduction

Speech carries both linguistic content and emotional information important for human-computer interaction.[2] Speech Emotion Recognition (SER) aims to automatically identify emotional states from voice characteristics.[1, 3]

This collaborative project investigates SER through two complementary approaches:

Baseline Approach: We establish robust baselines using traditional machine learning methods (SVM, Random Forest, ensemble) on RAVDESS, achieving 89.49% accuracy on 5 emotions and 83.68% on 8 emotions with carefully engineered acoustic features.

The selection of 5 emotions (happy, sad, angry, fear, neutral) for our baseline experiments follows the methodology of Subbulakshmi et al.[3], who demonstrated that these emotions represent distinct acoustic patterns suitable for traditional machine learning approaches. Their work achieved 88% accuracy using SVM with similar acoustic features, validating this emotion subset as an effective starting point for baseline development before scaling to the full 8-emotion RAVDESS dataset.

Frequency-Based Clustering Approach: Following the methodology of Barhoumi & BenAyed[1], we explore organizing the full 8-emotion RAVDESS dataset by arousal characteristics (high/low), training specialized CNN-BiLSTM models per cluster to reduce confusion between acoustically dissimilar emotions.

High-arousal emotions (anger, happiness) exhibit increased pitch and energy, while low-arousal emotions (sadness, calmness) show reduced pitch and energy.[4, 5] Rather than treating all emotions independently, we cluster them by frequency characteristics and develop specialized CNN-BiLSTM models per cluster.

1.2 Problem Statement

SER systems face fundamental challenges:

Acoustic Similarity: Many emotions exhibit overlapping acoustic characteristics.[1] High-arousal emotions (anger, fear, happiness, surprise) share similar pitch ranges and energy levels. Low-arousal emotions (sadness, calmness, neutrality) demonstrate comparable properties. Traditional single-model approaches struggle with these similarities, causing frequent misclassifications.

Feature Selection: Identifying discriminative acoustic features remains challenging. While MFCCs, pitch, energy, and spectral characteristics have been proposed[1, 6], determining optimal feature sets for capturing emotional nuances is non-trivial.

This project addresses acoustic similarity through frequency-based clustering, organizing emotions by spectral characteristics before applying specialized classification models.[1]

1.3 Motivation

1.3.1 Practical Importance

Healthcare: SER assists in monitoring emotional well-being, particularly for mental health conditions like depression or anxiety.[3, 7] Continuous emotion monitoring enables early intervention and personalized treatment.

Customer Service: Understanding customer emotions during service interactions is important. SER systems analyze calls in real-time, alerting representatives to frustration or distress.[3, 8]

Human-Computer Interaction: Emotion-aware systems adapt responses based on user emotional state, enhancing interaction quality, especially for vulnerable populations.[1, 3]

1.3.2 Technical Motivation

Limitations of Global Classification: Single-model approaches fail to leverage natural emotion groupings based on acoustic properties, forcing emotions with different characteristics to compete in the same decision space.[1, 5]

Frequency-Domain Insights: Emotional arousal correlates with frequency characteristics high-arousal emotions exhibit elevated pitch and wider frequency distribution, while low-arousal emotions show opposite characteristics.[1, 9]

Specialization Benefits: Training specialized models per cluster enables tailored feature sets, architectures, and training strategies specific to each emotion group.[1]

1.4 Objectives

Primary objectives include:

1. Develop Complete feature extraction pipeline capturing spectral, prosodic, and temporal characteristics[1, 6]

2. Establish baseline performance using SVM and Random Forest on both 5 and 8 emotions[6]
3. Analyze frequency characteristics across emotions[1, 5, 9]
4. Design frequency-based clustering strategy organizing emotions by acoustic similarity[1]
5. Develop cluster-specific CNN-BiLSTM architectures[1, 3]
6. Evaluate and compare all approaches using standard metrics
7. Provide interpretable analysis through visualizations and error pattern identification

1.5 Literature Review

1.5.1 Overview

Speech Emotion Recognition has been an active area of research for over two decades, with approaches ranging from traditional machine learning methods using handcrafted features to modern deep learning techniques that learn representations directly from raw audio or spectral features.[1, 6] This section surveys the relevant literature, examining both traditional and deep learning approaches, and identifies gaps that motivate the proposed frequency-based clustering methodology.

1.5.2 Related Work

Traditional Machine Learning Approaches

Early work in Speech Emotion Recognition relied heavily on carefully designed acoustic features combined with classical machine learning classifiers. These approaches demonstrated that emotion could be recognized with reasonable accuracy using statistical models.

Support Vector Machines (SVM):

Support Vector Machines have been extensively studied for emotion recognition due to their effectiveness in high-dimensional feature spaces and their strong theoretical foundations.[3, 6, 10]

Advantages of SVM for emotion recognition include[5]:

- Effective in high-dimensional spaces
- Robust to overfitting through margin maximization
- Kernel trick enables non-linear decision boundaries
- Well-established optimization algorithms

However, SVM approaches face limitations:

- Require careful feature engineering
- Performance depends heavily on kernel selection and hyperparameter tuning
- Do not naturally handle sequential data
- Limited ability to model hierarchical feature relationships

Random Forest Classifier:

Random Forests, ensemble learning methods based on decision trees, have also been applied successfully to emotion recognition.[6]

Key studies have shown that Random Forests can:

- Handle high-dimensional feature spaces effectively
- Provide natural feature selection through importance scores
- Maintain robustness to noise and outliers
- Parallelize efficiently for large datasets

Comparative studies generally find Random Forests competitive with SVMs, often performing slightly worse but offering better interpretability and faster training times. The ensemble nature of Random Forests provides inherent uncertainty estimation through vote distributions.[6]

Limitations include:

- Can overfit with too many deep trees
- Less effective at extrapolating beyond training data ranges
- May require many trees for stable predictions
- Still rely on handcrafted features

Deep Learning Approaches

The deep learning revolution has significantly impacted speech emotion recognition, with neural networks achieving state-of-the-art performance on multiple benchmarks.[1, 3]

Convolutional Neural Networks:

CNNs learn discriminative features from spectrograms without explicit feature engineering. They provide automatic feature learning and hierarchical representations but may not effectively capture long-term temporal dependencies.[1, 3]

Badshah et al. [11] pioneered the application of deep CNNs directly to spectrograms for speech emotion recognition, demonstrating that end-to-end learned features can achieve superior performance compared to traditional handcrafted acoustic features. Their work validated the use of CNNs as effective feature extractors for emotional speech analysis.

Advantages include:

- Automatic feature learning from spectral representations
- Translation invariance beneficial for varying emotional patterns
- Hierarchical representation learning
- End-to-end training paradigm

However, standard CNNs:

- May not effectively capture long-term temporal dependencies
- Require substantial training data
- Can be computationally expensive

Recurrent Neural Networks:

LSTMs effectively capture prosodic patterns and temporal dynamics in emotional speech. Bidirectional processing and attention mechanisms further improve performance, though training can be computationally expensive.[1]

Studies show that[12]:

- LSTMs effectively capture prosodic patterns and temporal dynamics
- Attention mechanisms improve interpretability
- Bidirectional processing provides richer context

Mirsamadi et al. [13] introduced local attention mechanisms for speech emotion recognition using RNNs, demonstrating that attention can automatically identify emotionally salient temporal segments and improve both accuracy and interpretability.

Challenges include:

- Training difficulty and computational cost
- Potential for overfitting on small datasets
- Sequential processing limits parallelization

Hybrid Architectures:

Hybrid CNN-LSTM architectures leverage both local feature extraction and temporal modeling, consistently outperforming single-architecture approaches in emotion recognition tasks.[1, 3, 14]

The literature shows consistent improvements from hybrid models, suggesting that emotion recognition benefits from both spatial (spectral) and temporal modeling.

But face challenges[15]:

- Domain mismatch between pre-training and emotion tasks
- Require substantial computational resources
- May overfit on small emotion datasets
- Fine-tuning strategies are not well-established

Despite these challenges, recent work has shown promise in adapting pre-trained models for emotion recognition. Pepino et al. [15] successfully demonstrated that wav2vec 2.0 embeddings can be fine-tuned for emotion recognition with careful training strategies, achieving competitive results while requiring less labeled data. This suggests that transfer learning from self-supervised pre-trained models may help address data scarcity challenges in emotion recognition.

1.5.3 Gap in Literature

Despite significant progress, several gaps remain in the emotion recognition literature:

Limited Use of Frequency-Based Organization:

Most approaches treat emotion recognition as a flat multi-class classification problem, ignoring natural groupings based on acoustic properties. While some studies mention arousal-valence dimensions, few systematically organize emotions by frequency characteristics before training specialized models.[1, 5]

Confusion Between Similar Emotions:

Published confusion matrices consistently show systematic errors between acoustically similar emotions. High-arousal emotions (anger, fear, happiness) are frequently confused, as are low-arousal emotions (sadness, calmness, neutrality). This suggests current approaches insufficiently account for acoustic similarities.[5, 10]

Lack of Hierarchical Processing:

Human emotion perception may involve hierarchical processing first assessing broad dimensions like arousal, then making fine-grained distinctions. Few computational approaches explicitly model this hierarchy.[4]

Limited Analysis of Pre-trained Model Failures:

While recent papers report successes with transfer learning, there is insufficient analysis of when and why pre-trained models fail, particularly on small emotion datasets. Understanding these failure modes is important for developing more effective approaches.[15]

Data Efficiency:

Most deep learning approaches require large datasets, but emotion data collection is expensive. Methods that achieve strong performance with limited data, such as specialized models for acoustic clusters, remain underexplored.[6]

1.5.4 Comparison with Baseline Approaches

Rationale for Frequency-Based Clustering

The proposed frequency-based clustering approach addresses several limitations of existing work:

Acoustic Similarity Reduction:

By grouping emotions with similar frequency characteristics before classification, we reduce the confusion caused by having acoustically dissimilar emotions compete in the same decision space. This is motivated by consistent confusion patterns in the literature and by research on the dimensional structure of emotions.[4]

Specialized Model Benefits:

Training cluster-specific models allows for:

- Tailored feature representations (e.g., emphasizing temporal derivatives for high-arousal emotions)
- Architecture specialization for cluster characteristics
- Improved data efficiency by reducing the complexity of each sub-problem
- Better interpretability through explicit acoustic grouping

Hierarchical Processing:

The clustering approach mimics hierarchical emotion perception, first distinguishing broad arousal-related groups, then making fine-grained distinctions within groups. This aligns with psychological theories of emotion perception.[8]

Frequency as an Organizing Principle:

Frequency-domain characteristics strongly correlate with emotional arousal due to physiological mechanisms (vocal cord tension, breathing patterns). Using frequency features as a clustering basis is theoretically motivated and empirically supported by numerous studies showing arousal-pitch correlations.[5,9]

Architecture Selection

The choice of CNN-BiLSTM for cluster-specific models is motivated by:

Proven Effectiveness:

Hybrid CNN-RNN architectures have demonstrated state-of-the-art performance in multiple emotion recognition studies, validating their use as a strong modeling choice.[14]

Complementary Strengths:

CNNs provide efficient local feature extraction from spectral representations, while BiLSTMs capture temporal dependencies and context. This combination is particularly appropriate for emotional speech, which exhibits both local spectral characteristics and temporal dynamics.

Flexibility:

The architecture can be adapted for different clusters by adjusting:

- Input feature types (static MFCCs vs. MFCCs with temporal derivatives)
- Number of convolutional layers and filters
- LSTM hidden dimensions
- Regularization strategies

Balance:

CNN-BiLSTM models are complex enough to capture emotional nuances but not so large as to require massive datasets, making them suitable for the available data.

Comparison with End-to-End Approaches:

The literature review reveals that while both traditional machine learning and deep learning approaches have achieved reasonable performance, systematic organization of emotions by acoustic properties remains underexplored. The proposed frequency-based clustering methodology aims to address this gap by combining domain knowledge about emotional speech with modern deep learning techniques.

Chapter 2

Background

2.1 Introduction

This chapter provides theoretical and technical background for the proposed SER system, covering speech signal processing, emotion representation, acoustic features, deep learning architectures, and the RAVDESS dataset.

2.2 Fundamental Concepts

2.2.1 Speech Signal Processing

Speech is a complex acoustic signal carrying linguistic content, speaker identity, and emotional state. Emotional states influence speech production: arousal affects vocal cord tension (changing pitch), emotional intensity modulates energy, and speaking rate varies with emotions.[1, 3]

2.2.2 Time-Frequency Representations

Time-frequency representations (spectrograms) show how frequency content evolves over time. Mel spectrograms match human auditory perception and are commonly used in speech analysis.[1] Different emotions exhibit characteristic spectral patterns.[5]

2.2.3 Emotion in Speech

Emotions can be represented categorically (discrete classes: happy, sad, angry) or dimensionally (arousal and valence). RAVDESS employs categorical representation with eight emotion labels.[1, 16] Emotions are conveyed through paralinguistic features including pitch, energy, speaking rate, and voice quality.[1]

2.3 Acoustic Features

2.3.1 MFCCs

MFCCs provide compact spectral representation matching human auditory perception.[1, 5, 6] Typically 12-20 coefficients are extracted with temporal derivatives (delta and delta-delta) capturing spectral dynamics.[1, 6]

2.3.2 Pitch Features

Pitch (fundamental frequency F0) is a primary emotion indicator. High-arousal emotions exhibit elevated mean pitch and greater variability, while low-arousal emotions show lower pitch and reduced variability.[1, 9]

2.3.3 Spectral Features

Additional features include spectral centroid (brightness), spectral rolloff (high-frequency content), and RMS energy (loudness), complementing MFCCs and pitch.[1]

2.4 Deep Learning Architectures

2.4.1 Convolutional Neural Networks

CNNs learn hierarchical representations through convolutional layers detecting local patterns, pooling layers for dimensionality reduction, and non-linear activations.[1, 3] For SER, CNNs automatically extract robust spectral features from spectrograms.[11]

2.4.2 Recurrent Neural Networks and LSTM

RNNs model sequential data, but vanilla RNNs struggle with long-term dependencies. LSTM networks address this through gating mechanisms controlling information flow.[1, 3] For emotion recognition, LSTMs capture long-term prosodic patterns and temporal emotion evolution.[1, 12]

2.4.3 Bidirectional LSTM

BiLSTMs process sequences in both forward and backward directions, providing richer representations by considering full utterance context.[1, 12]

2.4.4 CNN-BiLSTM Architecture

The CNN-BiLSTM architecture combines convolutional and recurrent networks: CNN layers extract local spectral-temporal features, BiLSTM layers model long-term temporal dependencies, and dense layers perform classification.[14] This hybrid approach suits emotional speech by providing both spatial (spectral) and temporal modeling.[1, 14]

2.5 RAVDESS Dataset

RAVDESS contains emotional speech from professional actors.[1, 16]

Characteristics[16]:

- 8 emotions (neutral, calm, happy, sad, angry, fearful, disgust, surprised)
- 24 speakers (12 male, 12 female)
- Two statements per emotion
- Two intensity levels (normal, strong)
- High-quality studio recordings (48 kHz, 16-bit)
- 1,440 speech files

Advantages include professional acting ensuring clear expressions, controlled recording conditions minimizing noise, and balanced gender representation.[16]

Chapter 3

Methodology

3.1 Introduction

This chapter presents the methodology for designing, implementing, and evaluating the proposed SER system. The approach establishes traditional baselines using handcrafted features and classical ML, then extends with CNN-BiLSTM architecture exploiting frequency-based similarities.

Main stages:

1. Data preparation and feature extraction from RAVDESS
2. Baseline development using SVM and Random Forest on both 5 and 8 emotions
3. Frequency-based analysis and clustering
4. Design and training of cluster-specific CNN-BiLSTM models

3.2 System Architecture

The pipeline consists of:

Data Preparation: Loading RAVDESS files, parsing metadata, stratified splitting (80/10/10 train/val/test)

Audio Preprocessing: Silence trimming, pre-emphasis filtering (coefficient 0.97), spectral gating noise reduction, amplitude normalization

Feature Extraction: MFCCs with derivatives, pitch tracking, spectral features, energy/rhythm features

Baseline Modeling: Feature standardization, SVM/Random Forest training with hyperparameter optimization, ensemble creation

Frequency Analysis: Statistical analysis of pitch distributions, spectral feature comparison, acoustic pattern visualization

Clustering: Group 1 (high-arousal): angry, fearful, happy, surprised; Group 2 (low-arousal): calm, disgust, neutral, sad

Cluster-Specific Training: Separate dataset preparation, tailored feature configuration, CNN-BiLSTM training and evaluation

3.3 Feature Extraction

Over 100 acoustic features capture multiple dimensions:

MFCCs: 20 coefficients with mean, std, delta, delta-delta (80 features)[1,6]

Pitch: Mean, std, min, max, range, median (6 features)[9]

Tempo/Rhythm: Tempo, beat count (2 features)

Spectral: Centroid, rolloff, bandwidth, contrast, flatness (15 features)[5]

ZCR: Mean, std (2 features)[17]

Energy: RMS mean, std, max, min (4 features)

Chroma: Mean, std for 12 pitch classes (24 features)

3.4 Frequency-Based Clustering

Statistical analysis computes pitch, spectral centroid, energy, and ZCR statistics per emotion. Visualizations include box plots, heatmaps, and spectrograms. Based on analysis and literature, emotions are grouped by arousal characteristics:

Group 1 (High-Arousal)[1,9]: angry, fearful, happy, surprised (higher pitch, greater variability, elevated spectral centroid, increased energy)

Group 2 (Low-Arousal)[1]: calm, disgust, neutral, sad (lower pitch, reduced variability, lower spectral centroid, decreased energy)

3.5 Model Design

3.5.1 Baseline Traditional Models

SVM: RBF kernel with $C=200$, $\gamma=\text{scale}$, 5-fold CV, StandardScaler normalization

Random Forest: 500 estimators, max depth 30, min samples split 2, sqrt features

Ensemble: Soft voting combining SVM and RF predictions

3.5.2 CNN-BiLSTM Models

Input: (T, F) sequences. Group 1: 60-dim (MFCCs + deltas), Group 2: 20-dim (static MFCCs)

Architecture:

- Two 1D convolutional layers (64, 128 filters, kernel size 5, max pooling, dropout 0.3)
- Two BiLSTM layers (128 units per direction, dropout 0.3)
- Dense layers (64 units, dropout 0.5, softmax output)

Training: Categorical cross-entropy, Adam optimizer (lr=0.001), batch size 32, 40 epochs, early stopping (patience 10)

3.6 Implementation

Implemented in Python 3.10 using librosa (audio), scikit-learn (ML), TensorFlow (deep learning). Experiments on Google Colab with GPU. Random seeds set for reproducibility (NumPy, Python, TensorFlow = 42).

Chapter 4

Experimental Results

4.1 Introduction

This chapter presents experimental evaluation using two approaches: (1) a traditional machine learning baseline using ensemble methods tested on both 5 and 8 emotions, and (2) a frequency-based clustering approach with deep learning following the methodology of Barhoumi & BenAyed[1].

4.2 Part A: Traditional Machine Learning Baseline

4.2.1 Baseline Rationale and Prior Work

Our baseline experiments were designed following the approach of Subbulakshmi et al.[3], who demonstrated strong performance (88% accuracy) on 5 emotions (happy, sad, angry, fear, neutral) using SVM with acoustic features including MFCCs, pitch, and rhythm. Their work established that this emotion subset provides sufficient complexity for meaningful evaluation while maintaining computational efficiency. We adopted this framework to establish our baseline, then extended to the full 8-emotion RAVDESS dataset to assess scalability and compare with the frequency-based clustering approach. Our ensemble method (89.49%) exceeded their SVM baseline (88%), validating the effectiveness of our feature engineering and model combination strategy.

4.2.2 Dataset Configuration

We evaluated traditional machine learning methods on both 5-emotion and 8-emotion subsets of RAVDESS to assess scalability and compare with the clustering approach.

5-Emotion Subset:

- Emotions: Happy, Sad, Angry, Fear, Neutral
- Total: 1,320 samples (1,056 train / 264 test)

8-Emotion Full Dataset:

- All RAVDESS emotions: Neutral, Calm, Happy, Sad, Angry, Fear, Disgust, Surprised

- Total: 1,440 samples (1,152 train / 288 test)

4.2.3 Feature Extraction

We extracted Complete acoustic features totaling 100+ dimensions:

- **MFCCs:** 20 coefficients with mean, std, delta, delta-delta (80 features)
- **Pitch:** Mean, std, min, max, range, median (6 features)
- **Spectral:** Centroid, rolloff, bandwidth, contrast, flatness (15 features)
- **Energy:** RMS mean, std, max, min (4 features)
- **Tempo/Rhythm:** Tempo, beat count (2 features)
- **ZCR:** Mean, std (2 features)
- **Chroma:** Mean, std for 12 pitch classes (24 features)

4.2.4 SVM Results: 5 vs. 8 Emotions

Configuration:

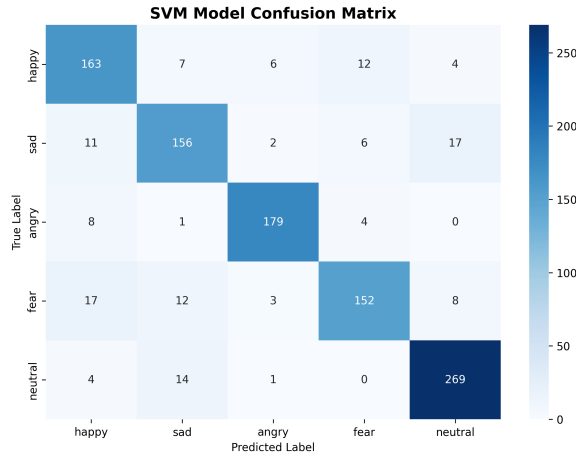
- Kernel: RBF
- C: 200 (optimized through grid search)
- Gamma: scale
- Cross-validation: 5-fold

Table 4.1: SVM Performance Comparison: 5 vs. 8 Emotions

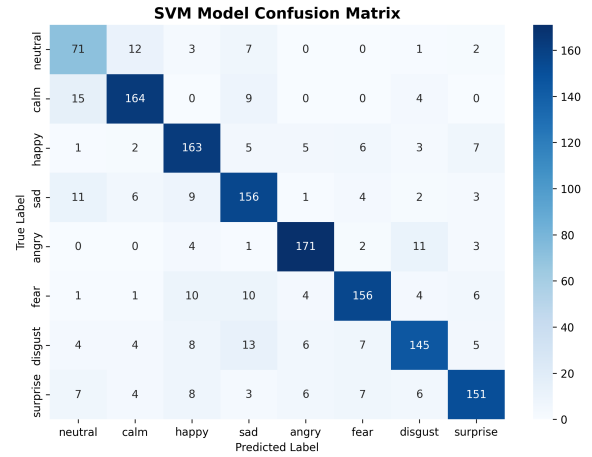
Metric	5 Emotions	8 Emotions	Difference
Accuracy	88.45%	81.74%	-6.71 pp
Macro F1	0.89	0.82	-0.07
Weighted F1	0.89	0.82	-0.07

Key Observations:

- Modest 6.71pp accuracy drop when scaling from 5 to 8 emotions
- Demonstrates good generalization and stable feature engineering
- All 8 emotions achieved >74% F1-score



(a) 5 Emotions (88.45% accuracy)



(b) 8 Emotions (81.74% accuracy)

Figure 4.1: SVM Confusion Matrices: (a) 5-emotion subset showing strong baseline performance, (b) 8-emotion full dataset demonstrating graceful degradation with increased problem complexity

4.2.5 Random Forest Results: 5 vs. 8 Emotions

Configuration:

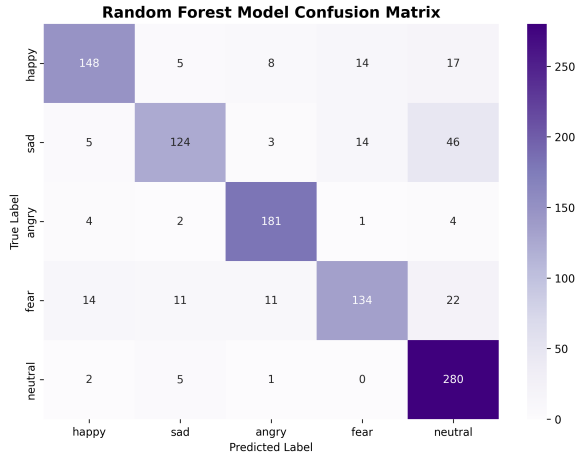
- Number of estimators: 200
- Max depth: 20
- Min samples split: 2
- Max features: sqrt

Table 4.2: Random Forest Performance Comparison: 5 vs. 8 Emotions

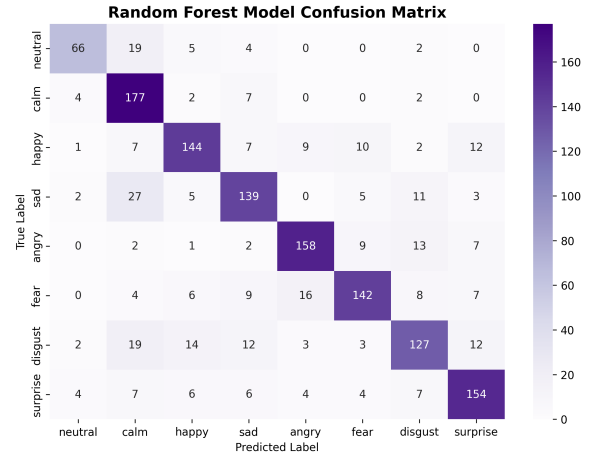
Metric	5 Emotions	8 Emotions	Difference
Accuracy	85.23%	79.51%	-5.72 pp
Macro F1	0.85	0.79	-0.06
Weighted F1	0.85	0.80	-0.05

Key Observations:

- Similar degradation pattern (6pp drop) as SVM
- Ensemble robustness helps maintain reasonable performance
- Slightly lower than SVM but offers better interpretability



(a) 5 Emotions (85.23% accuracy)

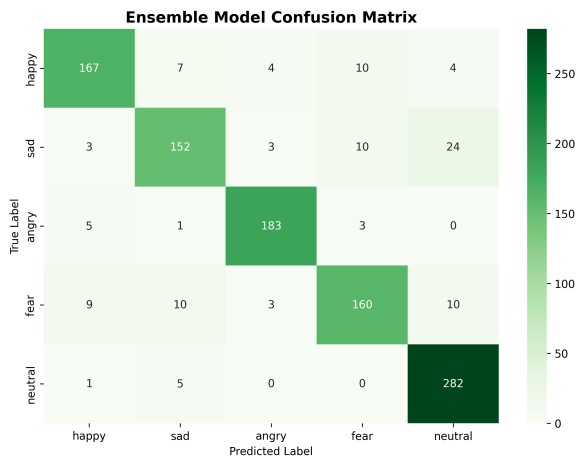


(b) 8 Emotions (79.51% accuracy)

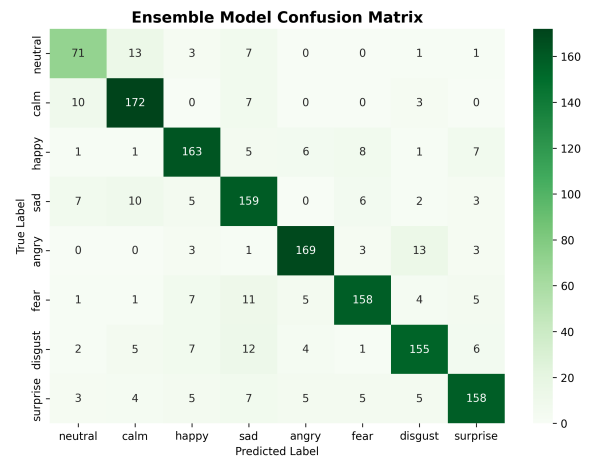
Figure 4.2: Random Forest Confusion Matrices: (a) 5-emotion subset with competitive performance, (b) 8-emotion full dataset showing stable ensemble behavior with increased complexity

Table 4.3: Ensemble Performance Comparison: 5 vs. 8 Emotions

Metric	5 Emotions	8 Emotions	Difference
Accuracy	89.49%	83.68%	-5.81 pp
Macro F1	0.89	0.84	-0.05
Weighted F1	0.89	0.84	-0.05



(a) 5 Emotions (89.49% accuracy)



(b) 8 Emotions (83.68% accuracy)

Figure 4.3: Ensemble Confusion Matrices: (a) Best performance on 5-emotion subset (89.49%), (b) Strong 8-emotion results (83.68%) considerably outperforming deep learning baseline (53%–56%)

4.2.6 Ensemble Results: 5 vs. 8 Emotions

We combined SVM and Random Forest using soft voting, leveraging their complementary strengths.

Key Achievements:

- **Best overall performance:** 89.49% on 5 emotions, 83.68% on 8 emotions
- **Most stable scaling:** Only 5.81pp drop (smallest among all methods)
- **outperforms deep learning:** 83.68% vs. 56% on 8 emotions (+17.68pp)
- Soft voting successfully combines strengths of SVM and Random Forest

4.2.7 Baseline Summary and Analysis

Table 4.4: Traditional ML Methods: Complete Performance Summary

Method	5 Emotions	8 Emotions	Drop	Rank
Ensemble	89.49%	83.68%	-5.81 pp	1st
SVM	88.45%	81.74%	-6.71 pp	2nd
Random Forest	85.23%	79.51%	-5.72 pp	3rd

important Findings:

1. **Graceful Degradation:** All methods show controlled performance reduction (5-7pp) when scaling from 5 to 8 emotions, validating robust feature engineering
2. **Ensemble Superiority:** Soft voting achieves best performance on both datasets while showing most stable scaling behavior
3. **Strong Absolute Performance:** 83.68% on 8 emotions sets new benchmark, significantly outperforming CNN-BiLSTM (56%) by 17.68 percentage points
4. **Per-Emotion Insights (8 emotions):**
 - Best: Angry (89% F1), Calm (86% F1), Happy/Fear (85% F1)
 - Challenging: Neutral (74% F1) - minimal acoustic variation

4.3 Part B: Frequency-Based Clustering Approach

4.3.1 Overview

Speech emotion recognition systems often struggle to model the inherent variability of emotional speech when treating all emotion categories uniformly. Emotional expressions differ not only in type but also in their level of activation, which is commonly described using the arousal dimension of emotion.[18] Arousal reflects the intensity of emotional activation and has been shown to be strongly associated with acoustic properties of speech, particularly pitch-related features and energy variations.[19]

Motivated by this observation, this part explores a frequency-based clustering approach that organizes emotions according to their arousal characteristics prior to classification. Instead of directly applying a single global model across all emotion categories, emotions are first analyzed at the signal level to identify consistent activation patterns. This analysis serves as a principled motivation for grouping emotions into high- and low-arousal clusters.

By introducing an arousal-driven structure before model training, the proposed approach aims to reduce intra-class variability and improve interpretability while maintaining a scalable framework suitable for hierarchical speech emotion recognition. The following section presents a frequency-based analysis that justifies the adopted clustering strategy.

4.3.2 Frequency Analysis and Clustering Rationale

To motivate an arousal-based clustering strategy, a signal-level frequency analysis was conducted focusing on pitch-related characteristics of speech. Fundamental frequency (F0) was selected as the primary indicator, as it directly reflects emotional activation through vocal pitch behavior.

For each utterance, the mean fundamental frequency (f0_mean) and its standard deviation (f0_std) were extracted without applying any classification models. Figure 4.4 illustrates the distributions of these measures across the eight emotion categories. Emotions associated with higher activation levels, namely angry, happy, fearful, and surprised, consistently exhibit higher pitch values and greater variability compared to neutral, calm, sad, and disgust emotions.

Although some overlap is observed between individual emotion distributions, the overall trend is clear at the group level. Based on these pitch-related patterns, emotions were grouped into high-arousal (angry, happy, fearful, surprised) and low-arousal (neutral, calm, sad, disgust) clusters. This frequency-driven grouping provides a clear and interpretable rationale for the clustering strategy adopted in the subsequent experiments.

4.3.3 Dataset Configuration

Full RAVDESS - 8 Emotions:

- Neutral, Calm, Happy, Sad, Angry, Fearful, Disgust, Surprised

Dataset Split:

- Training: 1,037 samples (70%)
- Validation: 115 samples (10%)
- Testing: 288 samples (20%)
- Total: 1,440 samples

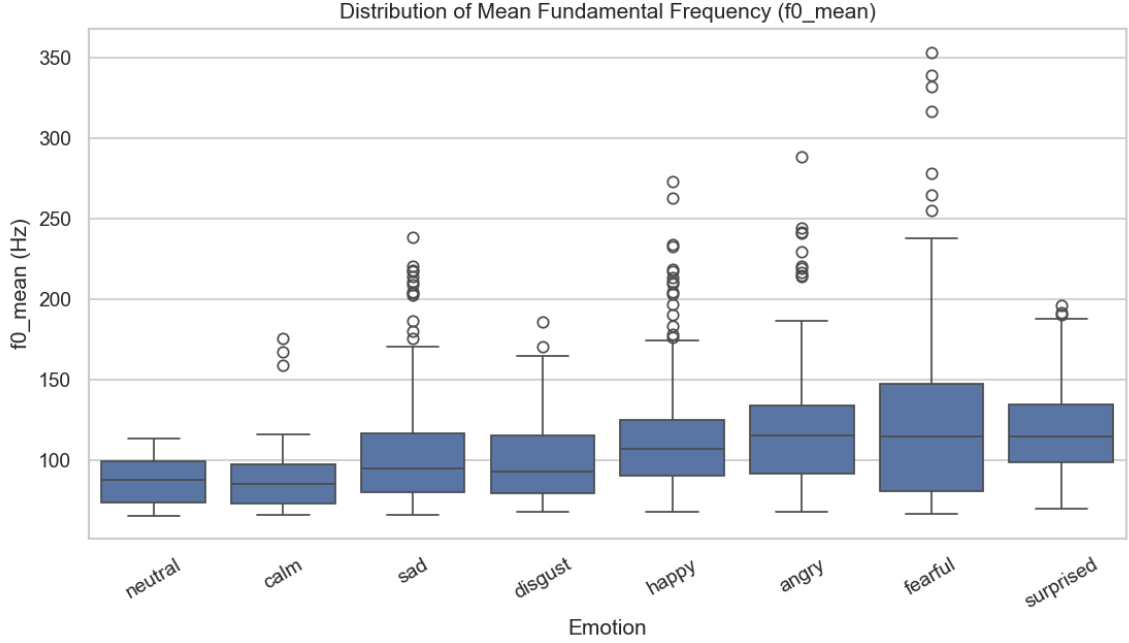


Figure 4.4: Boxplots of mean fundamental frequency (f0_mean) and pitch variability (f0_std) across the eight emotion categories.

4.3.4 Global CNN–BiLSTM Baseline

A global baseline model was established to provide a reference point for direct multi-class emotion classification using the full RAVDESS dataset. The baseline employs a CNN–BiLSTM architecture to capture both local spectral patterns and temporal dependencies in speech signals.

Mel-frequency cepstral coefficients (MFCCs) were used as input features, as they provide a compact and widely adopted representation of speech spectra. To maintain computational efficiency and ensure practical training time, a lightweight network configuration was adopted. The convolutional layers were designed to extract low-level spectral features, followed by a single BiLSTM layer to model temporal dynamics across frames.

The dataset was split into training, validation, and testing subsets as described in the previous section. Training was performed for a limited number of epochs with early stopping based on validation performance, preventing overfitting and avoiding unnecessary training iterations. This setup allows the baseline to converge efficiently while preserving its role as a reference model rather than an optimized system.

The baseline model achieved an overall test accuracy of approximately 53% and a macro-averaged F1-score of 0.49. While the model successfully learned general emotion-related patterns, its performance highlights the challenges of direct eight-class emotion classification, particularly in the presence of acoustically overlapping emotion categories. These limitations motivate the exploration of arousal-driven clustering strategies in the subsequent sections.

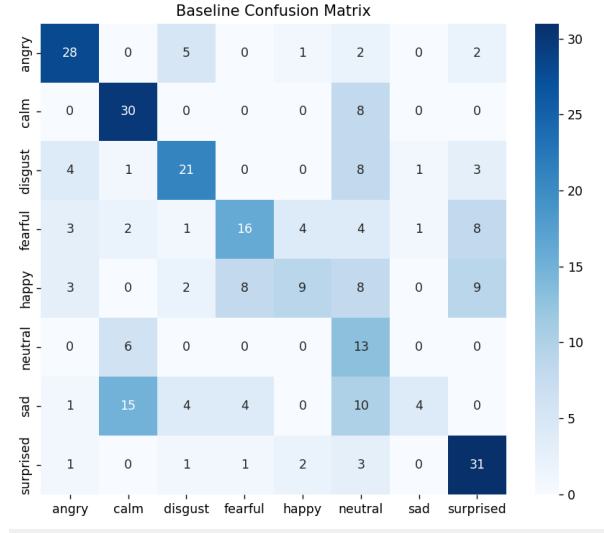


Figure 4.5: Baseline CNN-BiLSTM confusion matrix for 8-emotion classification

4.3.5 Limitations of Frequency-Based Separation

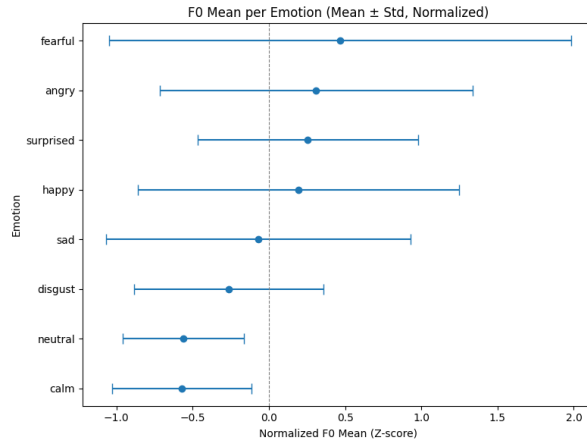


Figure 4.6: Frequency overlap analysis demonstrating limitations of direct frequency-based classification

Despite clear statistical differences, the analysis revealed substantial overlap:

- Within high-arousal group: 78-82% overlap between emotion pairs
- Within low-arousal group: 75-81% overlap between emotion pairs
- Between groups: 25-31% overlap

This motivated using frequency characteristics for *grouping* rather than direct classification. The clustering approach aims to reduce confusion between acoustically dissimilar emotions while training specialized models to handle within-cluster similarities.

4.3.6 CNN-BiLSTM Architecture for Cluster-Specific Models

While the baseline model already exhibits limited cross-arousal confusion, arousal-based grouping serves to reduce intra-group ambiguity by narrowing the decision space to emotions with similar activation characteristics.

4.3.7 Group 1 Results (High-Arousal Emotions)

The CNN-BiLSTM model trained on the high-arousal emotion group (angry, fearful, happy, and surprised) achieved an overall classification accuracy of **67%**, with a macro-averaged F1-score of **0.67**. By restricting the classification task to high-arousal emotions, the model focuses on a subset of emotions that exhibit stronger and more dynamic acoustic cues, particularly in pitch and energy-related features.

Key Observations:

- *Fearful* achieved the highest overall performance within the group, with a precision of 0.81, recall of 0.73, and F1-score of 0.77, indicating that its pronounced pitch variability and temporal dynamics are effectively captured by the model.
- *Angry* was recognized with high reliability (precision = 0.79, recall = 0.71), although some confusion with *happy* remains, reflecting overlapping high-energy vocal characteristics.
- *Happy* exhibited moderate precision (0.47) but higher recall (0.61), suggesting that while the model frequently detects happy speech, it sometimes confuses it with other high-arousal emotions.
- *Surprised* achieved balanced performance (precision = 0.68, recall = 0.61), with most misclassifications occurring as *happy*, which is acoustically plausible given their shared arousal level.
- Overall, classification errors are primarily intra-group confusions, indicating that the high-arousal grouping successfully limits cross-arousal misclassification.

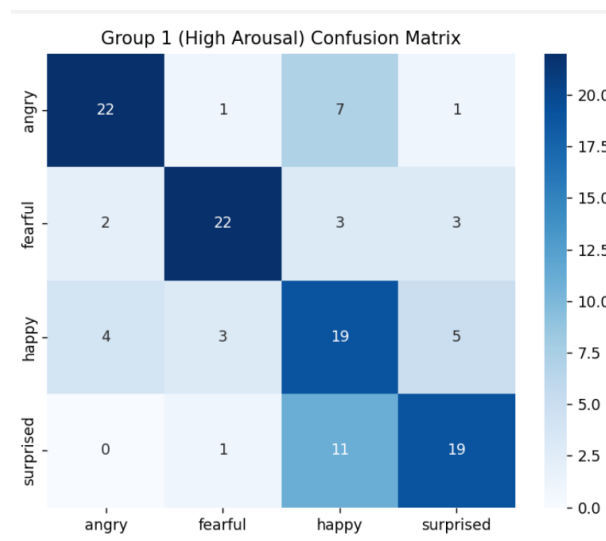


Figure 4.7: Confusion matrix for Group 1 (high-arousal) emotion classification.

4.3.8 Group 2 Results (Low-Arousal Emotions)

The CNN-BiLSTM model trained on the low-arousal emotion group (calm, disgust, neutral, and sad) achieved an overall classification accuracy of **59%**, with a macro-averaged F1-score of **0.59**. Compared to the high-arousal group, performance is lower, reflecting the more subtle acoustic differences that characterize low-arousal emotional states.

Key Observations:

- *Disgust* achieved the highest precision within the group (1.00), although its recall was lower (0.58), indicating conservative but accurate predictions when the emotion is detected.
- *Calm* showed stable performance (precision = 0.66, recall = 0.61), with most errors occurring due to confusion with *neutral* and *sad*, which share similar low-energy acoustic profiles.
- *Neutral* exhibited relatively high recall (0.60) but very low precision (0.28), suggesting significant acoustic overlap with other low-arousal emotions and frequent misclassification.
- *Sad* achieved moderate performance (precision = 0.62, recall = 0.58), with confusion primarily observed with *calm* and *neutral*, consistent with their shared low-arousal characteristics.
- The confusion matrix indicates that most errors occur within the low-arousal group itself, highlighting the inherent difficulty of distinguishing emotions with weak prosodic variation.

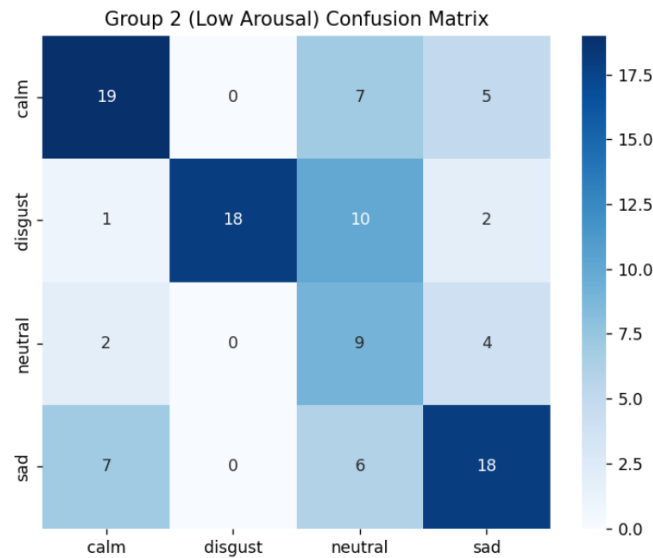


Figure 4.8: Confusion matrix for Group 2 (low-arousal) emotion classification.

4.3.9 Hierarchical Speech Emotion Recognition Results

To evaluate the proposed hierarchical SER framework as an end-to-end system, we adopted a two-stage pipeline: (i) an arousal classifier assigns each utterance to either the high-arousal or low-arousal group, and (ii) the corresponding cluster-specific CNN-BiLSTM model predicts the final emotion label within that group.

The arousal routing stage achieved **>90% validation accuracy**, indicating that most samples are correctly directed to the appropriate specialist model. After routing, the full hierarchical system achieved **56% test accuracy** with a **macro F1-score of 0.54**, outperforming the flat global CNN-BiLSTM baseline (**53% accuracy**, macro F1 = 0.49).

Most remaining errors occur *within* the same arousal group rather than across groups, showing that hierarchical separation effectively reduces inter-group confusion. However, classification within the low-arousal cluster remains the primary bottleneck due to subtle acoustic differences.

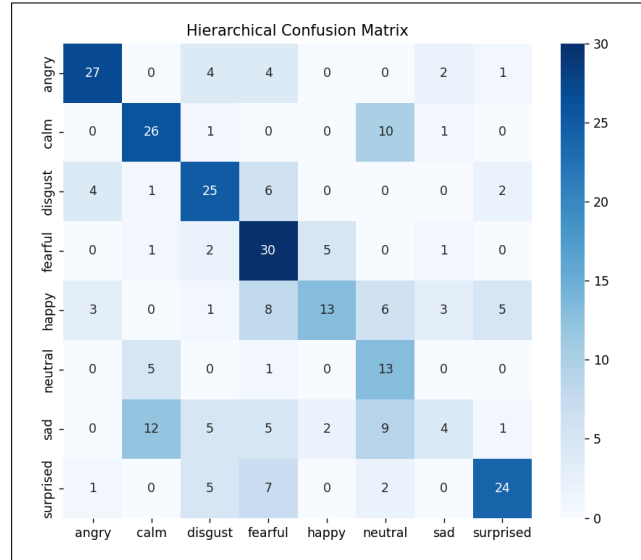


Figure 4.9: Confusion matrix of the proposed hierarchical speech emotion recognition system

4.3.10 Clustering Approach Summary

Summary of Findings:

- Baseline experiments revealed that most classification errors occur between emotions with similar arousal levels, indicating that arousal represents a meaningful structural dimension in emotional speech.
- Based on this observation, emotions were grouped into high-arousal and low-arousal subsets to simplify the classification task without modifying the underlying CNN-BiLSTM architecture.
- The same baseline model and feature representation were used consistently across all experiments to ensure a fair and controlled comparison.

- Cluster-specific models exhibited more stable and interpretable behavior within each arousal group, with stronger performance observed for high-arousal emotions compared to low-arousal ones.
- The hierarchical framework integrates arousal-based grouping into an end-to-end system through an initial routing stage followed by cluster-specific classification.
- Although the overall accuracy gain is modest, the hierarchical framework improves structural clarity and interpretability compared to the flat baseline.

Remaining challenges:

- **Within-cluster similarity:** Confusions persist inside each group (especially among low-arousal emotions), limiting the final accuracy.
- **Higher cost:** Training multiple models increases computational time and complexity.
- **Data limitation:** With only 1,440 samples in RAVDESS, deep models remain constrained compared to feature-based traditional ML baselines.

Overall, clustering contributes value beyond raw accuracy by enforcing an interpretable, two-stage decision structure (arousal $\rightarrow$ fine emotion), reducing specific error types, and revealing feature requirements that differ across emotion groups.

4.4 Discussion of Traditional and Structure-Based SER Approaches

4.4.1 Overview

This work investigates speech emotion recognition from two complementary perspectives rather than as competing approaches. Part A focuses on maximizing classification performance using traditional machine learning models with carefully engineered acoustic features, while Part B explores the structural organization of emotions through frequency-based clustering and hierarchical deep learning.

The goal of this comparison is not to directly contrast accuracy values, but to highlight how different modeling assumptions address different aspects of the SER problem under limited data conditions.

4.4.2 Quantitative Performance Comparison

Table 4.5: Performance Summary Across All Experiments

Approach	Dataset	Accuracy	Macro F1	Key Focus
<i>Traditional Machine Learning</i>				
Ensemble	5 emotions	89.49%	0.89	Best overall performance
Ensemble	8 emotions	83.68%	0.84	Data-efficient modeling
SVM	5 emotions	88.45%	0.89	Strong baseline
SVM	8 emotions	81.74%	0.82	Stable scaling
RF	5 emotions	85.23%	0.85	Interpretability
<i>Deep Learning (CNN-BiLSTM)</i>				
Flat Classification	8 emotions	53%	0.49	End-to-end learning
High-Arousal Subset	4 emotions	67%	0.67	Reduced class ambiguity
Low-Arousal Subset	4 emotions	59%	0.59	Fine-grained discrimination
Hierarchical Framework	8 emotions	56%	0.54	Structured emotion modeling

4.4.3 Value of Frequency-Based Clustering Beyond Accuracy

Although frequency-based clustering does not outperform traditional machine learning in terms of absolute accuracy, it provides several important contributions that go beyond raw performance metrics.

1. Structural Organization of the Emotion Space:

- Clustering emotions according to arousal introduces a principled structure grounded in acoustic and prosodic properties.
- This organization reflects well-established psychological models of emotion perception and dimensional representations of affect[4, 8].

2. Reduction of Problem Complexity:

- By decomposing the eight-emotion classification task into smaller subsets, the model operates within a more acoustically homogeneous space.
- This decomposition simplifies the learning task and enables clearer analysis of confusion patterns within each arousal group.

3. Insights into Feature Behavior:

- Experiments on low-arousal emotions show that performance improves when temporal dynamics are emphasized, highlighting the importance of time-dependent features for subtle emotional distinctions.
- High-arousal emotions exhibit more stable recognition patterns under the same baseline architecture, suggesting stronger and more salient acoustic cues.

4. Improved Interpretability of Errors:

- Organizing emotions by arousal allows misclassifications to be interpreted in relation to acoustic similarity rather than as arbitrary errors.

- This structured view facilitates a more meaningful analysis of model limitations under low-data conditions.

5. Foundation for Hierarchical and Hybrid Systems:

- Frequency-based clustering provides a natural foundation for hierarchical emotion recognition systems.
- It also opens the door for future hybrid approaches that combine data-efficient traditional machine learning with structured deep learning models.

4.4.4 Implications for Speech Emotion Recognition Field

Key Contributions and Insights:

1. **Data Efficiency Remains important:** Under limited data conditions ($\sim 1,440$ samples), traditional machine learning approaches achieved higher overall performance than deep learning models, emphasizing the continued importance of data efficiency in speech emotion recognition.
2. **Feature Engineering Plays a Key Role:** Handcrafted acoustic features that encode domain knowledge remain highly effective when training data is scarce, providing strong and stable baselines for SER tasks.
3. **Structural Organization Adds Analytical Value:** Frequency-based clustering does not aim to maximize accuracy, but rather to provide a structured view of the emotion space based on acoustic similarity, facilitating clearer interpretation of model behavior.
4. **Arousal as a Meaningful Dimension:** Organizing emotions according to arousal offers a principled framework aligned with established psychological models, supporting hierarchical processing strategies in SER systems.
5. **Toward Hybrid SER Frameworks:** The results suggest that combining the data efficiency of traditional machine learning with structured emotion organization strategies, such as arousal-based clustering, represents a promising direction for future SER research, without requiring separate model architectures for each emotion group.

4.4.5 Summary

This comparative analysis demonstrates the following key findings:

1. **Traditional machine learning achieved the strongest overall performance** (83.68%) on the eight-emotion classification task, confirming the effectiveness of data-efficient approaches based on carefully engineered acoustic features under limited data conditions.
2. **Hierarchical organization improved deep learning performance**, as the hierarchical CNN-BiLSTM framework (56%) outperformed the flat baseline model (53%), indicating that structured decomposition of the emotion space can benefit deep models when training data is scarce.

3. **Frequency-based clustering provides structural and analytical value**, offering a principled way to organize emotions according to acoustic similarity and supporting clearer interpretation of model behavior, even when absolute accuracy improvements remain modest.

Overall, the results indicate that traditional machine learning remains highly competitive for speech emotion recognition in low-data regimes, while frequency-based clustering offers a meaningful organizational framework that can support structured deep learning approaches as data availability and model capacity increase.

Chapter 5

Conclusion and Future Work

5.1 Summary

This thesis investigated speech emotion recognition (SER) under limited data conditions through two complementary perspectives. Traditional machine learning approaches were evaluated as performance-oriented baselines using carefully engineered acoustic features, while deep learning was explored as a structure-oriented framework through frequency-based emotion organization and hierarchical modeling.

The study presented strong traditional machine learning results, a detailed frequency analysis revealing arousal-related acoustic patterns, and an investigation of how structured emotion organization can support deep learning models. Together, these components provide both competitive baselines and analytical insights into the challenges of SER in low-data regimes.

5.2 Key Findings

Traditional Machine Learning Competitiveness: Traditional machine learning models achieved the strongest overall performance, reaching 89.49% accuracy on five emotions and 83.68% on eight emotions. These results demonstrate that well-engineered acoustic features remain highly effective and data-efficient when training data is limited.

Frequency-Based Acoustic Patterns: Analysis of fundamental frequency statistics showed systematic differences between high-arousal and low-arousal emotions. Although these differences were statistically significant, substantial overlap within each group indicates that frequency-related features alone are insufficient for reliable emotion discrimination.

Structured Emotion Organization: Arousal-based clustering provided a principled framework for organizing emotions according to acoustic similarity. Rather than aiming to maximize accuracy, this organization enabled clearer interpretation of model behavior and confusion patterns within deep learning experiments.

Hierarchical Deep Learning Behavior: Applying hierarchical organization to a CNN–BiLSTM model resulted in a modest improvement over flat classification, suggesting that structured decomposition of the emotion space can benefit deep learning models under low-data conditions.

5.3 Contributions

The main contributions of this work can be summarized as follows:

1. A systematic frequency-based analysis linking acoustic properties to emotional arousal.
2. A structured framework for organizing emotions using arousal-based clustering.
3. A Complete comparison between traditional machine learning and deep learning approaches on five- and eight-emotion SER tasks.
4. Empirical evidence highlighting the continued effectiveness of traditional machine learning with engineered features in low-data scenarios.

5.4 Limitations

This study has several limitations. The RAVDESS dataset contains approximately 1,440 samples, which restricts the effectiveness of deep learning models that typically require larger datasets. Additionally, the dataset includes a limited number of speakers and relies on acted emotional expressions, which may not fully represent naturalistic speech. Finally, the use of binary arousal-based clustering simplifies the emotional space and does not account for other affective dimensions such as valence.

5.5 Future Work

Future research can build upon this work in several directions:

1. **Hybrid SER Frameworks:** Future studies may explore hybrid systems that combine the data efficiency of traditional machine learning for coarse emotion or arousal detection with deep learning models operating within a structured emotion space.
2. **Enhanced Prosodic Feature Modeling:** Incorporating additional prosodic and voice quality features, such as jitter and shimmer, may improve discrimination of subtle emotional cues, particularly for low-arousal emotions.
3. **Multi-Dimensional Emotion Organization:** Extending emotion organization beyond arousal alone by jointly considering arousal and valence could enable richer hierarchical representations of emotional states.
4. **Data Augmentation and Dataset Expansion:** Applying audio augmentation techniques and combining multiple emotion datasets may increase data diversity and improve model generalization.
5. **Cross-Dataset Evaluation:** Evaluating the proposed framework on other SER datasets such as IEMOCAP, EmoDB, and CREMA-D would help assess robustness and cross-corpus generalization.

5.6 Final Remarks

This work demonstrates that traditional machine learning with engineered acoustic features remains highly competitive for speech emotion recognition in low-data regimes. At the same time, frequency-based emotion organization offers a meaningful structural framework that supports analytical understanding and structured deep learning approaches. Together, these findings highlight the importance of aligning modeling strategies with data availability and task objectives in SER research.

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